

Quiz II

Algorithms Analysis and Design
IIT Hyderabad, Monsoon 2025

October 27, 2025

Maximum Marks: 30. Time: 45 min

There are 3 questions 10 marks each.

- Suppose that in addition to edge capacities, a flow network has vertex capacities. That is each vertex v has a limit $\ell(v)$ on how much flow can pass through v . Show how to transform a flow network $G = (V, E)$ with vertex capacities into an equivalent flow network $G' = (V', E')$ without vertex capacities, such that a maximum flow in G' has the same value as a maximum flow in G .
- The vertex-connectivity (analogous to edge connectivity) of an undirected graph is defined as the minimum number k of vertices that must be removed to disconnect the graph. Assuming that edge connectivity can be computed using maximum flow, can you also compute the vertex-connectivity (say, using vertex capacities etc.)? 7 + 3 = 10 marks
- 2. • Give an efficient greedy algorithm that finds an optimal vertex cover for a tree. Can you find a linear time algorithm for the same?
- Argue for (or against) how a max-flow algorithm can be used to design a 2-approximate algorithm for minimum vertex cover problem. 6 + 4 = 10 marks
- 3. • Is the class P closed under intersection? What about the class NP ? Prove your answers (make suitable assumptions, if required). 1 + 2 = 3 marks
- Let $CLIQUE_i = \{ \langle G \rangle \mid G \text{ has a clique of size } i \}$. Prove that $CLIQUE_{1000} \in P$. Assuming that 3-SAT is NP-Complete, prove that $CLIQUE_{\lfloor \frac{n}{3} \rfloor}$ is NP-Complete where n is the number of vertices in graph G . Show that for any i : $CLIQUE_{i+1} \leq_p CLIQUE_i$. Spot the error in the following proof of $P = NP$ and suggest suitable modifications to the author to correct the error: Consider induction on the clique size k . The base case is $k = 1$, and clearly $CLIQUE_1 \in P$. Also, you just proved that if $CLIQUE_i \in P$ then $CLIQUE_{i+1} \in P$. From induction, we know: $\forall k \in \mathbb{N} \text{ } CLIQUE_k \in P$. Thus $P = NP$. 1 + 3 + 1 + 2 = 7 marks

ALL THE BEST!